

Project 7: Simulation of a Sheet Forming Process

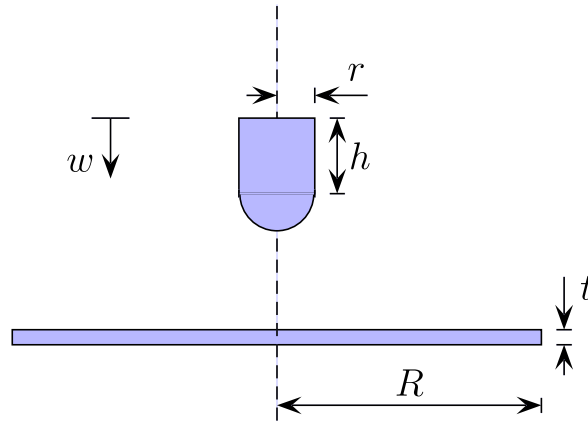


Abbildung 7.1: Geometry and boundary conditions.

The forming process of a circular sheet is simulated, see Fig. 7.1. While the sheet is subject to bilinear isotropic hardening, the stamp material is modeled as linear-elastic. The clamping of the sheet should be considered as sticking and is located in the radial area of $(r + t) < \text{radius} < R$. Due to the sticking of the clamping, it can be represented by suitable Dirichlet boundary conditions.

As for the contact modeling, the *Contact Wizard*-utility provided by ANSYS may be used.

- geometric data:
 $R = 250.0 \text{ mm}$ $t = 5.0 \text{ mm}$ $r = 50.0 \text{ mm}$ $h = 100.0 \text{ mm}$
- material data:
 - sheet
 $E = 3000 \text{ N/mm}^2$ $\nu = 0.2$ $\sigma_f = 60.0 \text{ N/mm}^2$ $T_f = 500 \text{ N/mm}^2$
 - stamp
 $E = 210,000 \text{ MPa}$ $\nu = 0.3$
- forming displacement:
 $\Delta w = 30 \text{ mm}$

Exercise:

A suitable boundary value problem should be established in ANSYS using the given information above. All task in ANSYS should be done using an APDL-script.

General Aspects:

- choose an appropriate finite element type and material model
- convergence study to estimate the necessary mesh density
- analysis of results

- comparison with analytical solution (if possible)
- all plots and graphs should be produced directly in ANSYS using APDL (not necessary diagrams related to convergence study)

Project-related Aspects:

- The stress-strain curve corresponding to the given material data has to be created.
- The overall motion should be animated (*.avi-file)
- Plots showing the contact pressure and penetration need to be created.
- The resulting plastic deformation of the sheet should be displayed.
- A force-displacement diagram for the stamp has to be generated.

Submission:

Following documents should be submitted only online

until September, 6th 2022, 12:00 p.m. !!!

!! one compressed archive including following files should be uploaded to moodle !!

- presentation (file format: pdf)
- documentation/report (file format: pdf)
- complete APDL input scripting (file format: plain text file e.g. *.txt)
- text file including project number, names of group members, student ids (file format: plain text file e.g. *.txt)
- animation of deformation process (file format: avi-file)

Presentation/Colloquium:

The project and its results should be documented in a report and also illustrated in a presentation (15-20 min) as part of the colloquium. Following aspects should be considered:

- title slide including names and student-ids of all group members and project number,
- boundary value problem (physical problem),
- modeling, boundary conditions, symmetry conditions, etc.
- aspects of the ANSYS-APDL-scripting (z. B. pseudocode, flow chart),
- results of project (see general and project-related aspects),
- conclusion and outlook.

Following this, a colloquium regarding the project and the advanced processing in ANSYS will take place.

NOTE: Please follow the general guideline of good scientific practice. Especially, take correct citations and references of sources into account.